

What's Your Automation Style: Brainy, Brawny or Tiny?



Janani Gopalakrishnan Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

In July this year, Changying Precision Technology Co., a manufacturer of mobile phone modules, set up the first fully-automated factory at Dongguan, China. Any time during the day or night, you can see 60 robot arms at ten production lines polishing the modules, a job that required 650 workers in the past. According to reports, 90 per cent of the factory's workforce comprises robots, with just a few human workers to monitor the machinery.

Elsewhere in the world, Ports of Auckland plans to replace 50 jobs with Autostrads, which are 15m-tall automated straddle carriers. These container-stacking robots can stack up to four containers high and are expected to increase the container terminal capacity by 30 per cent.

The job for a robot does not necessarily have to be high-precision, as in the first case, or heavy-load as in the second, it could just be a routine job that needs to be done well.

A wet-wipe maker, for example, recently signed up NEXCOM to design an automated system with Delta robots and EtherCAT master stations to coordinate everything from programmable logic controllers (PLCs), conveyors and sensors to gluing and labelling equipment. This move is expected to reduce manpower by around 30 per cent, while increasing productivity up to 1.5 packages per second.

Robots are being increasingly employed to automate innumerable tasks at manufacturing facilities across the world, not to forget their newfound jobs at homes, to make life easier and more comfortable. This article looks at the same trend, from angles that you might not have noticed before.

Coat-wearing robots

Known as laboratory robots, smart and efficient bots lend a helping hand

in laboratories, too. These help in jobs like high throughput screening, chemical and biochemical synthesis, so that researchers' time is freed for more strategic planning.

In a recent *Laboratory Equipment* story, Sam Michael, director of automation and compound management at National Centre for Advancing Translational Sciences (NCATS), USA, explains how the trend has shifted from large, monolithic robots to little robotic arms that have a special direct drive or magnetic drive.

While the size has shrunk, their importance has increased manifold, especially at NCATS, which is a heavily-automated research lab that runs high throughput screening of compounds. It recently hit the news as Yale University researchers used the NCATS facility to discover that a drug, originally meant for cancer, could restore memory and reverse cognitive problems in mice with Alzheimer-type symptoms.

The Machine. An example of a machine that can quickly solve a problem is called, well, The Machine, which was developed by Dr Martin Burke of University of Illinois, Urbana-Champaign. His goal was to automate chemical synthesis, so that several Ph.D-level scientists could be freed from working for days in the lab to develop small molecules. The device developed by him, currently licensed to Revolution Medicines Inc., uses a building-block approach to try and replicate what is found in nature.

The Machine can assemble complex small molecules at the click of a mouse. The idea is to break down complex molecules into smaller, chemical-building blocks, all of which have the same connector piece and can be stitched together with a single reaction. Burke's team automated this process, making The Machine work much like a 3D printer at the molecular level. They devised a simple catch-and-release method that adds one building block at a time, rins-

Fanuc CR-35iA (Image courtesy: www.controlsdrivesautomation.com)

